

A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

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Introduction

- The Hessian matrix is a target of interest, due to its high computational cost and its versatile applications.
- Atomistic Features, based on GFN2-xTB quantities, are used for the prediction of the target: Hessian matrix elements
- The features are permutation and rotational invariant to achieve a unique mapping onto the target.
- The zero point vibrational energy is used to quantify the precision of the model.

Overview of the ML Model

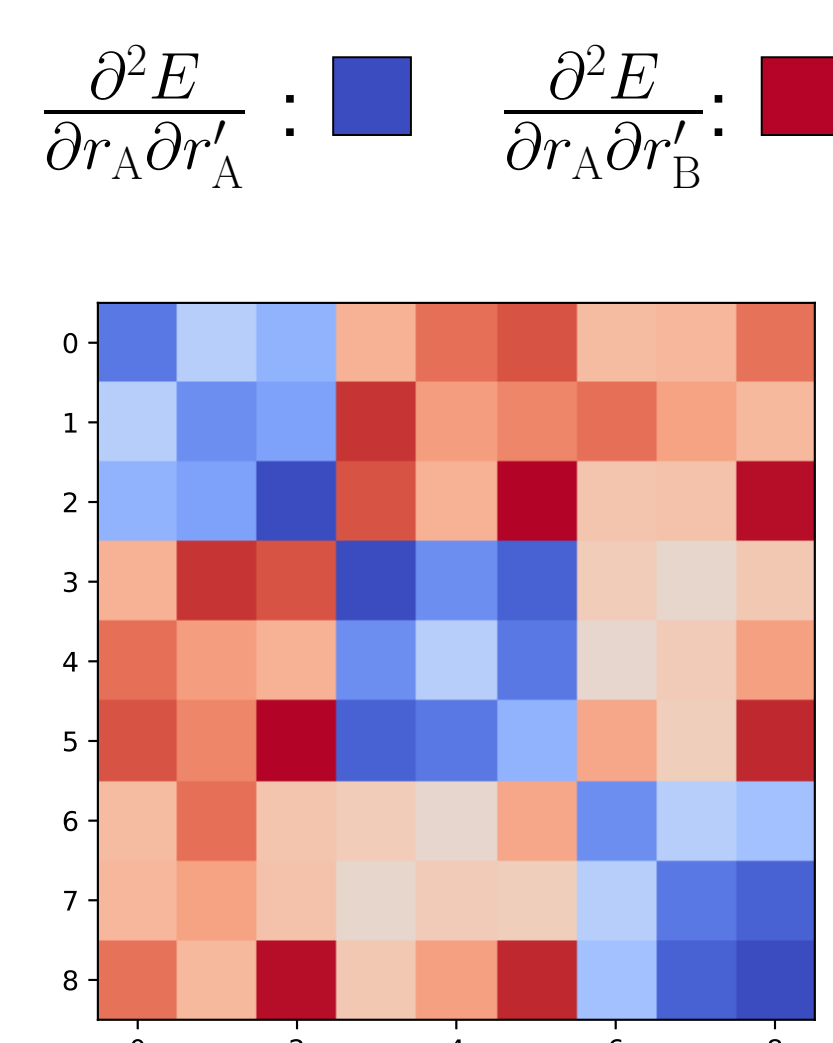
Origin of the Features

- Physics based, atomwise partitioned GFN2-xTB quantities are used for the features and were developed in our group and can be categorized into:

1. Geometry: e.g. Coordination Number
2. Density: e.g. Dipole Moment
3. Energy: e.g. Coulomb-repulsion energy

The Hessian Matrix

- The elements of the Hessian are derivatives either in respect of the coordinates of a single atom or of two atoms
- For the ML model, we distinguish between the diagonal blocks (blue) and off-diagonal blocks (red)

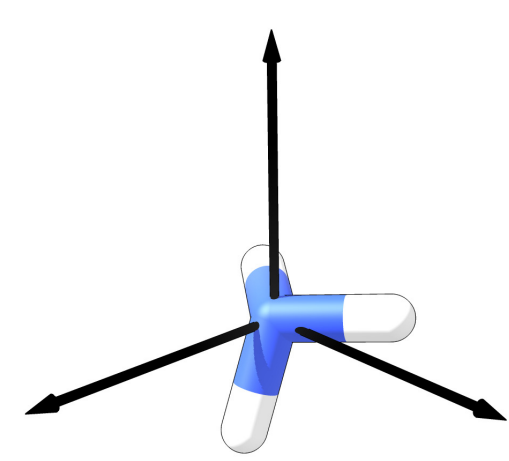
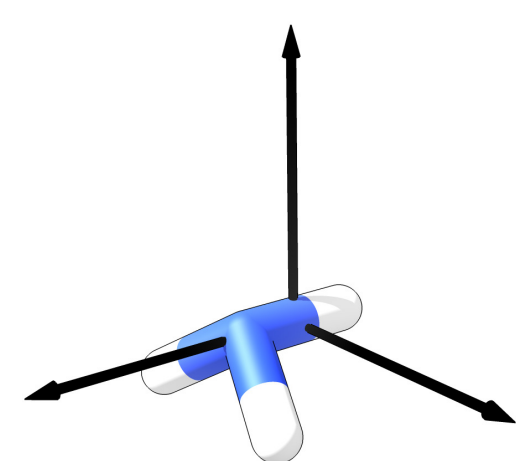


Connecting Features and Target

- The feature vector is either build up from the quantities of one or two atoms
- For ML modeling a constant feature vector length is necessary, hence two submodels are used
- The features and targets are computed for each 3×3 block in a different orientation to achieve a consistent orientation

Model  : Here the origin is shifted to the center of two atoms and the atoms are aligned to the z-axis.

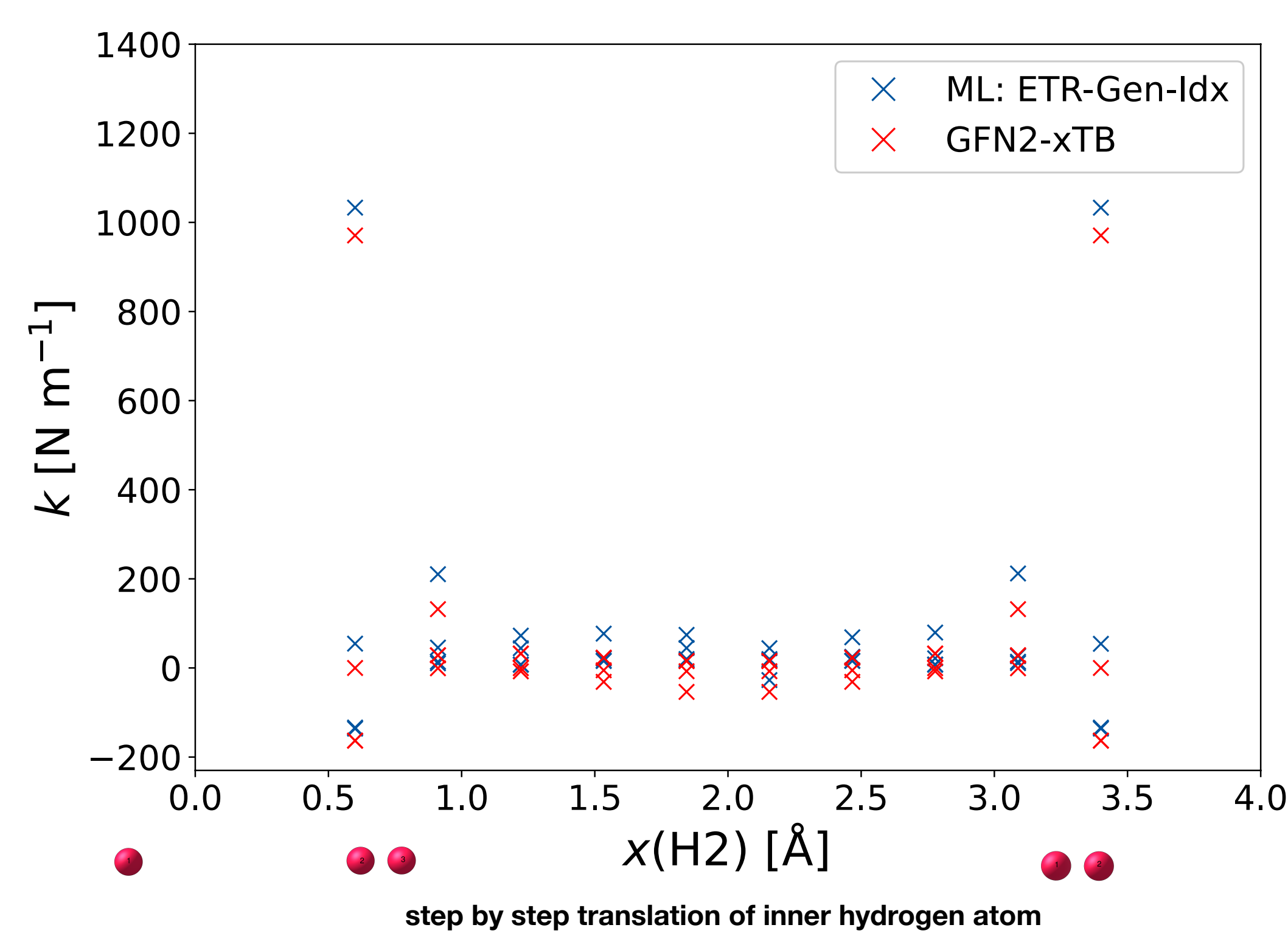
Model  : The origin of the coordinate system is shifted to the atom.



Unifying the Mapping of the Features

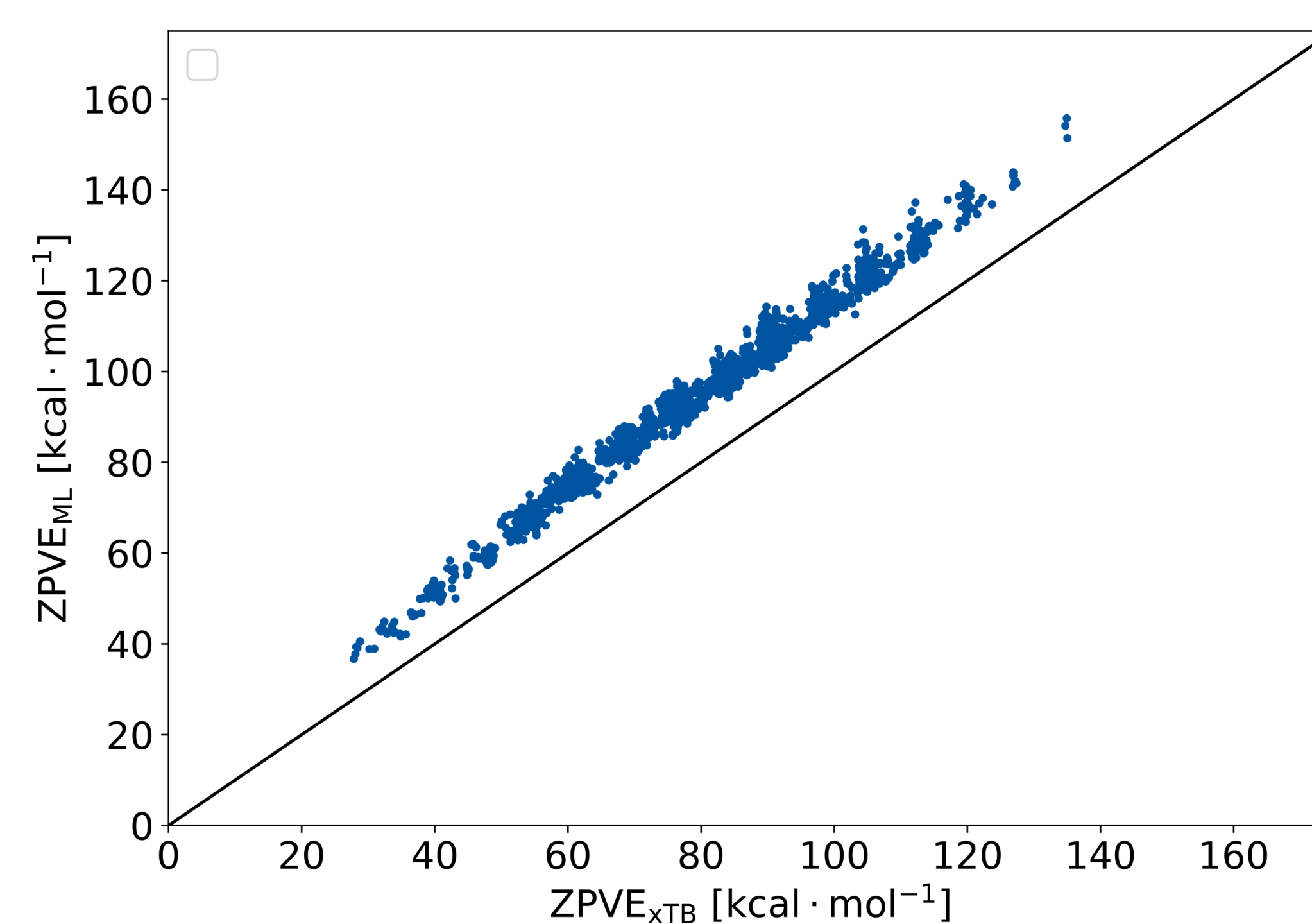
- An index is introduced to differ between the derivatives in respect to different coordinates of the same atom
- It is taken care of a consistent alignment to the atom oriented coordinate system

Verification of Unique Mapping



Predicted ZPVE of GBD7 Set

- Learning plots were computed using the GDB7 set, which consists of 7164 structures
- The test set was fixed to 25 % of the size of the GBD7 set
- The training set was increased up to 15% of the



Conclusion and Future Perspective

- Bullet point A
- Bullet point B
- Bullet point 3

References

- [1] A
- [2] B
- [3] C
- [4] D

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